Chapter 41

Campbell Biology, AP Edition, 12e

A herring gull eats a starfish.

Chapter 41

Animal Nutrition

Figure 41.1

For this herring gull, dinnertime has arrived. Once this meal has been taken in, the tissues of the sea star will be taken apart and its nutrients taken up. Paradoxically, the classes of nutrients in the sea star—largely proteins, fats, and carbohydrates—also make up the tissues of the gull.

How can animals extract the nutrients they need from food while not digesting their own tissues?

An animal digests food using compartmentalized processing in a tube-like system. The compartmentalized processing protects body tissues while allowing enzymes and acids to break down nutrients. Food goes into the system and secretions that promote digestion, especially enzymes break it down. Nutrients are transported through epithelial cells from the lumen and absorbed into the bloodstream. Digestive secretions catalyze breakdown of food and release of nutrients. Protective secretions help separate body cells from digestive activity. Waste goes out the end of the tube.

Key Concepts

- 41.1**An animal's diet must supply chemical energy, organic building blocks, and essential nutrients**
- **41.2Food processing involves ingestion, digestion, absorption, and elimination****
- **41.3Organs specialized for sequential stages of food processing form the mammalian digestive system****

- **41.4Evolutionary adaptations of vertebrate digestive systems correlate with diet****
- **41.5Feedback circuits regulate digestion, energy storage, and appetite****

Advanced placement edition

Cooperative interactions within the cells and systems of animals promote efficient use and exchange of energy and matter with the environment **(BIG IDEA 4)**, which all organisms need in order to grow, reproduce, and maintain organization **(BIG IDEA 2)**.

Study Tip

Make a flowchart: Digestion occurs stepwise in the human alimentary canal. As you study the molecular details, keep track of the overall process by adding to this start of a flowchart listing the overall effect and location of each step.

A flow chart of digestion.

The steps are as follows.

1. Break food into smaller pieces and mix with saliva (oral cavity).
2. Transfer boluses of food to stomach (esophagus).
3. Denature proteins in food (stomach).

Get Ready for This Chapter

In this chapter, you will learn how animals take in and digest food, move nutrients into cells, and allocate molecules for breakdown, biosynthesis, or storage. Before beginning this chapter, you should be able to do the following:

- Describe the structural organization of nucleic acids, proteins, polysaccharides, and fats (see Figure 5.6, Figure 5.9, Figure 5.15, and Figure 5.23).
- Distinguish between an acidic and a basic solution and between a hydrophobic and a hydrophilic molecule (see Concept 3.2).
- Outline the flow of energy through animal bodies (see Concept 9.1 and Figure 9.1).

The following questions provide a quick check of your basic knowledge in these areas.

1. Proteins consist of amino acids linked to

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phosphates.
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That's incorrect. Try again.

Incorrect. Review the following material: Proteins are a macromolecule constructed from one type of monomer linked into unbranched polymers by peptide bonds.

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sugars.
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That's incorrect. Try again.

Incorrect. Review the following material: Proteins are a macromolecule constructed from one type of monomer linked into unbranched polymers by peptide bonds.

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other amino acids.
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That's correct.

Correct! If you had trouble with this question, review the following material: Proteins are a macromolecule constructed from amino acids linked in unbranched polymers. The bond between amino acids is called a peptide bond, so a polymer of amino acids is called a polypeptide. Sugars, phosphates, and fatty acids are components of other macromolecules.

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fatty acids.
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That's incorrect. Try again.

Incorrect. Review the following material: Proteins are a macromolecule constructed from one type of monomer linked into unbranched polymers by peptide bonds.

2. Which of the following is hydrophobic and therefore insoluble in water?

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lipid
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That's correct.

Correct! If you had trouble with this question, review the following material: Lipids, which are nonpolar, are described as hydrophobic (from the Greek *phobos*, fearing) because they seem to repel water. An example from the kitchen is vegetable oil, which, as you likely know, does not mix stably with water-based substances such as vinegar. In contrast, phospholipids, nucleotides, and ribonucleotides are all polar and are soluble in water.

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phospholipid
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That's incorrect. Try again.

Incorrect. Review the following material: Nonpolar molecules tend not to mix stably with water-based substances and thus are hydrophobic.

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nucleotide
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That's incorrect. Try again.

Incorrect. Review the following material: Nonpolar molecules tend not to mix stably with water-based substances and thus are hydrophobic.

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ribonucleotide
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That's incorrect. Try again.

Incorrect. Review the following material: Nonpolar molecules tend not to mix stably with water-based substances and thus are hydrophobic.

3. Animals obtain the energy they need for growth by

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synthesizing macromolecules.
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That's incorrect. Try again.

Incorrect. Review the following material: Potential energy is stored in the bonds of complex organic molecules and can be captured when those bonds are broken.

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using ATP in food.
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That's incorrect. Try again.

Incorrect. Review the following material: Potential energy is stored in the bonds of complex organic molecules and can be captured when those bonds are broken.

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recycling heat from the environment.
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That's incorrect. Try again.

Incorrect. Review the following material: Potential energy is stored in the bonds of complex organic molecules and can be captured when those bonds are broken.

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breaking down organic molecules.
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That's correct.

Correct! If you had trouble with this question, review the following material: When enzymes in an animal's body break down complex organic molecules, some of those molecules' stored potential energy is captured; the rest is dissipated as heat that is lost to the environment. Often, the captured energy is stored in the form of ATP and is then used in synthesizing cell components, including macromolecules.

Learning Objectives

After studying this chapter, you should be able to do the following:

- 41.1Describe the dietary needs for chemical energy, organic building blocks, and the four classes of essential nutrients.
- 41.2Distinguish the main feeding mechanisms of animals, and define four stages of food processing.
- 41.3Describe the major functions of each compartment in the mammalian digestive system.
- 41.4Use examples to demonstrate how vertebrate digestive systems are adapted to diet.
- 41.5Provide examples of negative feedback in the regulation of nutrient intake, processing, and storage.

Concept 41.1: An animal's diet must supply chemical energy, organic building blocks, and essential nutrients

Although dining on sea stars, crabs, and fish is the herring gull's specialty, all animals eat other organisms—dead or alive, piecemeal or whole. Unlike plants, animals must consume food for both energy and the organic molecules used to assemble new molecules, cells, and tissues. Overall, an adequate diet must satisfy three needs: chemical energy for cellular processes, organic building blocks for macromolecules, and essential nutrients. The process by which an animal takes in and makes use of food to meet these needs constitutes **nutrition**.

All of an animal's activities depend on sources of chemical energy in the diet: carbohydrates, proteins, and lipids. The chemical reactions of cellular respiration use this energy to produce ATP, which powers processes ranging from DNA replication and cell division to vision and flight (see Concept 8.3). Ingested and digested nutrients can be broken down immediately or stored for future use.

In addition to fuel for ATP production, an animal requires raw materials for biosynthesis. To build the complex molecules it needs to grow, maintain itself, and reproduce, an animal's food must provide a source of organic carbon (such as sugar) and a source of organic nitrogen (such as protein).

The third requirement of an animal's diet is to provide **essential nutrients**, substances that an animal requires but cannot assemble from simple organic molecules.

Essential Nutrients

Essential nutrients in the diet include certain amino acids and fatty acids, as well as vitamins and minerals. The key functions of essential nutrients include serving as substrates of enzymes, as coenzymes, and as cofactors in biosynthetic reactions **(Figure 41.2)**.

Figure 41.2

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Roles of essential nutrients.
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The enzyme fatty acid desaturase catalyzes the reaction that converts linoleic acid to gamma-linoleic acid, a precursor for other cellular components. This reaction showcases the four classes of essential nutrients.

The enzyme fatty acid desaturase catalyzes the reaction that converts linoleic acid (an essential fatty acid with a hydrocarbon tail that is the substrate in this reaction) to gamma-linoleic acid. Gamma-linoleic acid is a precursor used in phospholipids (cell membrane components) and prostaglandins (used in cell signaling). This reaction showcases the four classes of essential nutrients, which are, 1, essential fatty acids; 2, vitamins (coenzyme); 3, minerals (cofactor); and 4, essential amino acids (monomers of polypeptides). The enzyme fatty acid desaturase contains the essential nutrients: vitamin B 3 (coenzyme), the mineral iron (cofactor), and a polypeptide chain consisting in parts of essential amino acids (monomers of polypeptides) including tyrosine, phenylalanine, leucine, and isoleucine.

This example of a biosynthetic reaction illustrates some common functions for essential nutrients. The conversion of linoleic acid to γ-linoleic acid by the enzyme fatty acid desaturase involves all four classes of essential nutrients, as labeled in blue. Note that almost all enzymes and other proteins in animals contain some essential amino acids, as indicated in the partial sequence shown for fatty acid desaturase.

In general, an animal can obtain all essential amino acids and fatty acids, as well as vitamins and minerals, by feeding on plants or other animals. Needs for particular nutrients vary among species. For instance, some animals (including humans) must get ascorbic acid (vitamin C) from their diet, whereas most animals can synthesize it from other nutrients.

Essential Amino Acids

All organisms require a standard set of 20 amino acids to make a complete set of proteins (see Figure 5.14). Plants and microorganisms normally can produce all 20. Most animals have the enzymes to synthesize about half of these amino acids, as long as their diet includes sulfur and organic nitrogen. The remaining amino acids must be obtained from the animal's food in prefabricated form and are therefore called **essential amino acids**. Many animals, including adult humans, require eight amino acids in their diet: isoleucine, leucine, lysine, methionine, phenylalanine, threonine, tryptophan, and valine. (Human infants also need a ninth, histidine.)

The proteins in animal products such as meat, eggs, and cheese are "complete," providing all the essential amino acids. In contrast, most plant proteins are "incomplete," being deficient in one or more essential amino acids. Corn (maize), for example, is deficient in tryptophan and lysine, whereas beans are lacking in methionine. However, vegetarians can easily obtain all of the essential amino acids by eating a varied diet of plant proteins.

Essential Fatty Acids

Animals require fatty acids to synthesize a variety of cellular components, including membrane phospholipids, signaling molecules, and storage fats. Although animals can synthesize many fatty acids, they lack the enzymes to form the double bonds found in certain required fatty acids. Instead, these molecules must be obtained from the diet and are considered **essential fatty acids**. In mammals, they include linoleic acid (see Figure 41.2). Animals typically obtain ample quantities of essential fatty acids from seeds, grains, and vegetables in their diet.

Vitamins

As Albert Szent-Györgyi, the discoverer of vitamin C, once quipped, "A vitamin is a substance that makes you ill if you *don't* eat it." **Vitamins** are organic molecules that are required in the diet in very small amounts (0.01–100 mg per day, depending on the vitamin).

The 13 vitamins required by humans vary in both chemical properties and function (**Table 41.1**). Vitamin B2, for example, is a water-soluble vitamin that is converted in the body to FAD, a coenzyme used in many metabolic processes, including cellular respiration (see Figure 9.12). Vitamin C, which is required for the production of connective tissue, is also water-soluble.

Table 41.1

Vitamin Requirements of Humans

A table lists the vitamin requirements of humans, including the major dietary sources, functions in the body, and symptoms of deficiency.

The table is divided into two sections as follows. Water soluble vitamins and fat-soluble vitamins.

Table 41.1 Vitamin requirements of Humans
Vitamin
Water-soluble vitamins
B 1, thiamine
B 2, riboflavin
B 3, niacin
B 5, pantothenic

B 6, pyridoxine
B 7, biotin
B 9, folic acid
B 12, cobalamin
C, ascorbic acid
Fat-soluble vitamins
A, retinol
D
E, tocopherol
K, phylloquinone

Fat-soluble vitamins include vitamin A, which is incorporated into visual pigments of the eye, and vitamin D, which aids in calcium absorption and bone formation. Our dietary requirement for vitamin D, unlike other vitamins, turns out to be variable. Why? When our skin is exposed to sunlight, our bodies synthesize vitamin D, reducing our dietary need.

For people with imbalanced diets, taking vitamin supplements at recommended daily levels is reasonable. It is far less clear that massive doses of vitamins confer any health benefits or are even safe. Moderate overdoses of water-soluble vitamins are probably harmless because excesses are excreted in urine. However, excesses of fat-soluble vitamins are deposited in body fat, so overconsumption may cause them to accumulate to toxic levels.

Minerals

Dietary **minerals** are inorganic nutrients, such as iron and sulfur, that are usually required in small amounts—from less than 1mg to about 2,500mg per day. As shown in **Table 41.2**, minerals have diverse functions in animal physiology. Some are assembled into the structure of proteins; iron, for example, is incorporated into the oxygen carrier hemoglobin as well as some enzymes (see Figure 41.2). Others, such as sodium, potassium, and chloride, are important in the functioning of nerves and muscles and in maintaining osmotic balance between cells and the surrounding body fluid. In vertebrates, the mineral iodine is incorporated into thyroid hormone, which regulates metabolic rate. Vertebrates also require relatively large quantities of calcium and phosphorus for building and maintaining bone.

Table 41.2

Mineral Requirements of Humans*

A table lists the mineral requirements of humans, including the major dietary sources, functions in the body, and symptoms of deficiency.

The table is as follows.

Table 41.2 Mineral requirements of Humans asterisk
Mineral
Calcium, C a
Phosphorous, P
Sulfur, S
Potassium, K
Chlorine, C I
Sodium, N a
Magnesium, M g
Iron, F e
Fluorine, F
Iodine, I
Asterisk Additional minerals required in trace amounts include Cobalt (C o), Copper (C u), manganese (M n), molybdenum (M o), Selenium (S e), and zinc (Z n). All of these minerals, as well as those in the table, can be harmful in excess. Note: following minerals required more than 200 milligrams per day: calcium, phosphorous, sulfur, potassium, chlorine, sodium, and magnesium.

Ingesting too much of some minerals can cause health problems. For example, excess sodium can contribute to high blood pressure. This is a particular problem in the United States, where the typical person consumes enough salt (sodium chloride) to provide about 20 times the required amount of sodium. Processed foods often contain large amounts of sodium chloride, even if they do not taste salty.

Variation in Diet

Despite many shared nutritional needs, animals have diverse diets. **Herbivores**, such as cattle, sea slugs, and caterpillars, dine mainly on plants or algae. **Carnivores**, such as sea otters, hawks, and spiders, mostly eat other animals. Bears, crows, cockroaches, and other **omnivores** (from the Latin *omnis*, all) don't in fact eat everything, but they do regularly consume animals as well as plants or algae. Humans are typically omnivores or herbivores.

The terms *herbivore, carnivore*, and *omnivore* represent the kinds of food an animal usually eats. However, most animals are opportunistic feeders, broadening their diet when their usual foods aren't available. For example, deer are herbivores, but occasionally eat insects, worms, or bird eggs. Similarly, herring gulls eat marine invertebrates, insects, and small fishes, but also human refuse. Note that microorganisms are an unavoidable "supplement" in every animal's diet.

Dietary Deficiencies

A diet that lacks one or more essential nutrients or consistently supplies less chemical energy than the body requires results in *malnutrition*, a failure to obtain adequate nutrition. Malnutrition affects one out of four children worldwide, impairing health and often survival.

Deficiencies in Essential Nutrients

Insufficient intake of essential nutrients can cause deformities, disease, and even death. For example, deer and other herbivores can develop fragile bones if the plants they consume grew in phosphorus-deficient soil. In such environments, some grazing animals obtain missing nutrients by consuming concentrated sources of salt or other minerals **(Figure 41.3)**. Similarly, some birds supplement their diet with snail shells, and certain tortoises obtain minerals from stones they ingest.

Figure 41.3

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Obtaining essential nutrients from an unusual source.
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Chamois kneels to lick salt.

A juvenile chamois (Rupicapra rupicapra), an herbivore, licks salts from exposed rocks in its alpine habitat. This behavior is common among herbivores that live where soils and plants provide insufficient amounts of minerals.

Like other animals, humans sometimes have diets lacking in essential nutrients. A diet with insufficient amounts of one or more essential amino acids causes protein deficiency, the most common type of malnutrition among humans. In children, protein deficiency may arise if their diet shifts entirely from breast milk to foods that contain relatively little protein, such as rice. Such children, if they survive infancy, often have impaired physical and mental development.

Undernourishment

As mentioned earlier, malnutrition can also be caused by a diet that fails to provide enough chemical energy. In this situation, the body first uses up stored carbohydrates and fat. It then begins breaking down its own proteins for fuel: Muscles shrink, and the brain may become protein-deficient. If energy intake remains less than energy expenditures, the animal will eventually die. Even if a seriously undernourished animal survives, some of the damage may be irreversible.

Inadequate nourishment in humans is most common when drought, war, or other crisis severely disrupts the food supply. The United Nations reported in 2021 that more than 34 million people globally were facing acute hunger, including many people in Yemen, South Sudan, northern Nigeria, Ethiopia, and Madagascar. In these and many other nations, conflict and severe climate events restricting the food supply have been further compounded by the COVID-19 pandemic.

Sometimes undernourishment occurs within well-fed human populations as a result of eating disorders. For example, anorexia nervosa involves weight loss to a level that is unhealthy for the individual's age and height and may be related to a distorted body image.

Assessing Nutritional Needs

Determining the ideal diet for the human population is an important but difficult problem for scientists. As objects of study, people present many challenges. Unlike laboratory animals, humans are genetically diverse. They also live in settings far more varied than the stable and uniform environment that scientists use in laboratory experiments. Ethical concerns present an additional barrier. For example, it is not acceptable to investigate the nutritional needs of children in a way that might harm a child's growth or development.

Many insights into human nutrition have come from *epidemiology*, the study of human health and disease at the population level. In the 1970s, for instance, researchers discovered that children born to women of low socioeconomic status were more likely to have neural tube defects, which occur when tissue fails to enclose the developing brain and spinal cord (see Concept 47.2). The English scientist Richard Smithells thought that malnutrition among these women might be responsible. As described in **Figure 41.4**, he found that vitamin supplementation greatly reduced the risk of neural tube defects. In other studies, he obtained

evidence that folic acid (vitamin B9) was the specific vitamin responsible, a finding confirmed by other researchers. Based on this evidence, the United States in 1998 began to require that folic acid be added to enriched grain products used to make bread, cereals, and other foods. Follow-up studies have documented the effectiveness of this program in reducing the frequency of neural tube defects. Thus, at a time when microsurgery and sophisticated diagnostic imaging dominate the headlines, a simple dietary change such as folic acid supplementation may be among the greatest contributors to human health.

Figure 41.4

Inquiry: Can diet influence the frequency of neural tube defects?

Experiment

Richard Smithells, of the University of Leeds, in England, examined the effect of vitamin supplementation on the risk of neural tube defects. People who had given birth to one or more babies with such a condition were put into two study groups. The experimental group consisted of those who were planning a pregnancy and began taking a multivitamin at least four weeks before attempting conception. The control group, who were not given vitamins, included those individuals who declined them and those who were already pregnant. The numbers of neural tube defects resulting from the pregnancies were recorded for each group.

Results

A table illustrates the vitamin supplements experimental and control groups for number of infants/fetuses studied and infants/ fetuses with aneural tube defect.

Data from R. W. Smithells et al., Possible prevention of neural-tube defects by periconceptional vitamin supplementation, *Lancet* 315:339–340 (1980).

Group	Number of Infants/Fetuses Studied	Infants/Fetuses with a Neural Tube Defect
Vitamin supplements (experimental group)	141	1

No vitamin supplements (control group)	204	12

Conclusion

This controlled study provided evidence that vitamin supplementation protects against neural tube defects, at least after the first pregnancy. Follow-up trials demonstrated that folic acid alone provided an equivalent protective effect.

INTERPRET THE DATA *After folic acid supplementation became standard in the United States, the frequency of neural tube defects dropped to an average of just one in* 5,000 *live births. Propose two explanations of why the observed frequency was much higher in the experimental group of the Smithells study.*

INQUIRY IN ACTION Read and analyze the original paper in *Inquiry in Action: Interpreting Scientific Papers*.

Concept 41.2: Food processing involves ingestion, digestion, absorption, and elimination

Whatever their diet, animals need to process food. We can divide food processing into ingestion, digestion, absorption, and elimination. The first stage—**ingestion**—is the act of eating or feeding. As shown in **Figure 41.5**, four quite different categories describe the feeding mechanisms of most animal species.

Figure 41.5

Exploring Four Main Feeding Mechanisms of Animals

Figure describes examples of four feeding mechanisms of animals: filter feeding, substrate feeding, fluid feeding, and bulk feeding

Filter feeding: Many aquatic animals are filter feeders, which strain small organisms or food particles from the surrounding medium. The humpback whale, shown above, is one example. Attached to the whale's upper

jaw are comb like plates called baleen, which remove small invertebrates and fish from enormous volumes of water and sometimes mud. Filter feeding in water is a type of suspension feeding, which also includes removing suspended food particles from the surrounding medium by capture or trapping mechanisms.

Substrate feeding: Substrate feeders are animals that live in or on their food source. This leaf miner caterpillar, the larva of a moth, is eating through the soft tissue of an oak leaf, leaving a dark trail of feces in its wake. Other substrate feeders include maggots (fly larvae), which burrow into animal carcasses.

Fluid feeding: Fluid feeders suck nutrient-rich fluid from a living host. This mosquito has pierced the skin of its human host with hollow, needlelike mouthparts and is consuming a blood meal (colorized SEM). Similarly, aphids are fluid feeders that tap the phloem sap of plants. In contrast to such parasites, some fluid feeders actually benefit their hosts. For example, hummingbirds and bees move pollen between flowers as they fluid-feed on nectar.

Bulk feeding: Most animals, including humans, are bulk feeders, which eat relatively large pieces of food. Their adaptations include tentacles, pincers, claws, venomous fangs, jaws, and teeth that kill their prey or tear off pieces of meat or vegetation. In this amazing scene, a rock python is beginning to ingest a gazelle it has captured and killed. Snakes cannot chew their food into pieces and must swallow it whole—even if the prey is much bigger than the diameter of the snake. They can do so because the lower jaw is loosely hinged to the skull by an elastic ligament that permits the mouth and throat to open very wide. After swallowing its prey, which may take more than an hour, the python will spend two weeks or longer digesting its meal.

During **digestion**, the second stage of food processing, food is broken down into molecules small enough for the body to absorb. Both mechanical and chemical processes are typically required. Mechanical digestion, such as chewing or grinding, breaks food into smaller pieces, increasing surface area. The food particles then undergo chemical digestion, which cleaves large molecules into smaller components.

Chemical digestion is necessary because animals cannot directly use the nucleic acids, fats, phospholipids, and most carbohydrates in food. These molecules are too large to pass through cell membranes and also are not all identical to those the animal needs for its particular tissues and functions. But when large molecules in food are broken down into their smaller components, the animal can use these products of digestion to assemble the large molecules it needs. For example, although the humpback whale and the tsetse fly in Figure 41.5 have very different diets, both break down proteins in their food to the same 20 amino acids from which they assemble all of the specific proteins in their bodies.

Enzyme-catalyzed synthesis of a fat or macromolecule links together smaller components, releasing a molecule of water for each new covalent bond formed. Chemical breakdown by digestive enzymes reverses this process, breaking bonds through the addition of water. This splitting process is called *enzymatic*

hydrolysis. Polysaccharides and disaccharides are split into simple sugars, as shown here for sucrose and the enzyme sucrase:

Enzymatic hydrolysis of a disaccharide

Sucrose ($C_{12}H_{22}O_{11}$) plus H_2O are split by sucrase, yielding glucose ($C_6H_{12}O_6$) plus fructose ($C_6H_{12}O_6$).

Similarly, proteins are broken down into small peptides and amino acids, and nucleic acids are cleaved into nucleotides and their components. Enzymatic hydrolysis also releases fatty acids and other components from fats and phospholipids. In many animals, bacteria living in the digestive system carry out some chemical digestion.

The last two stages of food processing occur after the food is digested. In the third stage, **absorption**, the animal's cells take up (absorb) small molecules such as amino acids and simple sugars. **Elimination**, in which undigested material passes out of the digestive system, completes the process.

Digestive Compartments

You have just read that digestive enzymes hydrolyze the same biological materials (such as proteins, fats, and carbohydrates) that make up the bodies of the animals themselves. How, then, are animals able to digest food without digesting their own cells and tissues? The evolutionary adaptation that allows animals to avoid self-digestion is the processing of food within specialized intracellular or extracellular compartments.

Intracellular Digestion

Food vacuoles—cellular organelles in which hydrolytic enzymes break down food—are the simplest digestive compartments. The hydrolysis of food inside vacuoles, called intracellular digestion, begins after a cell engulfs solid food by phagocytosis or liquid food by pinocytosis (see Figure 7.21). Newly formed food vacuoles fuse with lysosomes, organelles containing hydrolytic enzymes. This fusion of organelles brings food in contact with these enzymes, allowing digestion to occur safely within a compartment enclosed by a protective membrane. A few animals, such as sponges, digest all their food in this way (see Figure 33.4).

Extracellular Digestion

In most animal species, hydrolysis occurs largely by extracellular digestion, the breakdown of food in compartments that are continuous with the outside of the animal's body. Having one or more extracellular

compartments for digestion enables an animal to devour much larger pieces of food than can be ingested by phagocytosis.

Animals with relatively simple body plans typically have a digestive compartment with a single opening. This pouch, called a **gastrovascular cavity**, functions in digestion as well as in the distribution of nutrients throughout the body (hence the *vascular* part of the term). Small freshwater cnidarians called hydras provide a good example **(Figure 41.6)**. The hydra—a carnivore—uses its tentacles to stuff captured prey through its mouth into its gastrovascular cavity. Specialized gland cells of the hydra's gastrodermis, the tissue layer that lines the cavity, then secrete digestive enzymes that break the soft tissues of the prey into tiny pieces. Other cells of the gastrodermis engulf these food particles, and most of the hydrolysis of macromolecules occurs intracellularly. After the hydra has digested its meal, undigested materials that remain in its gastrovascular cavity, such as exoskeletons of small crustaceans, are eliminated through its mouth. Many flatworms also have a gastrovascular cavity (see Figure 33.9).

Figure 41.6

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Digestion in a hydra.
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Hydra digestion of Daphnia, includes digestive enzymes released from gland cells, followed by enzymatic breakdown of food into small particles, and then food particles engulfed and digested in food vacuoles.

Digestion begins in the gastrovascular cavity and is completed intracellularly after small food particles are engulfed by specialized cells of the gastrodermis.

Hydra digestion of Daphnia, a water flea. Hydra is a cnidarian with cylindrical body with a mouth surrounded by tentacles and a gastrovascular cavity. The gastrovascular cavity is lined by layers of cells including the outer epidermis and inner layer of gastrodermis. Digestion begins when the food is in the gastrovascular cavity.

1. Digestive enzymes are released by gland cells in the gastrodermis.
2. Enzymes break down food into small particles.
3. Food particles are engulfed by cells in the gastrodermis and digested in food vacuoles.

DRAW IT *Draw and label a simple diagram showing the pathway that nutrients follow from when food enters the hydra's mouth to when nutrients reach a cell on the outside of the tip of one of its tentacles.*

*For suggested answer, see*Appendix A.**

Video

Hydra Eating *Daphnia*

image

image

Rather than a gastrovascular cavity, animals with complex body plans have a digestive tube with two openings, a mouth and an anus **(Figure 41.7)**. Such a tube is called a *complete digestive tract* or, more commonly, an **alimentary canal**. Food moves along the alimentary canal in a single direction, encountering a series of specialized compartments that carry out stepwise digestion and nutrient absorption. An animal with an alimentary canal can ingest food while earlier meals are still being digested, a feat that is likely to be difficult or inefficient for an animal with a gastrovascular cavity.

Figure 41.7

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Variation in alimentary canals.
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Variations in alimentary canals.

These examples illustrate how the organization and structure of compartments for digestion, storage, and absorption differ among animals.

Three variations in three different species are as follows.

Part a. The alimentary canal of the earthworm, in order: mouth, pharynx, esophagus, crop, gizzard, intestine, anus. The alimentary canal of an earthworm includes a muscular pharynx that sucks food in through the mouth. Food passes through the esophagus and is stored and moistened in the crop. Mechanical digestion occurs in the muscular gizzard, which pulverizes food with the aid of small bits of sand and gravel. Further digestion and absorption occur in the intestine.

Part b. A grasshopper has several digestive chambers grouped into three main regions: a foregut, with an esophagus and crop; a midgut; and a hindgut. Food is moistened and stored in the crop, but most digestion occurs in the midgut. Pouches called gastric caeca (singular, ceca) extend from the beginning of the midgut and function in digestion and absorption.

Part c. Many birds have a crop for storing food and a stomach and gizzard for mechanically digesting it. Chemical digestion and absorption of nutrients occur in the intestine.

Because most animals have an alimentary canal, we'll use the complete digestive system of mammals in the next section to illustrate the general principles of food processing.

Concept 41.3: Organs specialized for sequential stages of food processing form the mammalian digestive system

In mammals, a number of accessory glands support food processing by secreting digestive juices through ducts into the alimentary canal. There are three pairs of salivary glands, as well as three individual glands: the pancreas, the liver, and the gallbladder. To explore the coordinated function of the accessory glands and alimentary canal, we'll consider the steps in food processing as a meal travels along the canal in a human.

The Oral Cavity, Pharynx, and Esophagus

As soon as food enters your mouth, or **oral cavity**, food processing begins **(Figure 41.8)**. Teeth with specialized shapes cut, mash, and grind, breaking the food into smaller pieces. This mechanical breakdown not only increases the surface area available for chemical breakdown but also facilitates swallowing. Meanwhile, the anticipation or arrival of food in the oral cavity triggers the release of saliva by the **salivary glands**.

Figure 41.8

The human digestive system.

The human digestive system. Structures are labeled and accessory glands are highlighted. The esophagus is a long, straight tube, while the intestines are heavily folded.

The human digestive system, described in order of the appearance of structures beginning with the mouth: mouth (with salivary glands highlighted as accessory glands); esophagus; stomach; small intestine (with accessory glands gall bladder, liver and pancreas highlighted); large intestine; rectum; anus.

After food is chewed and swallowed, it takes 5–10 seconds for it to pass down the esophagus and into the stomach, where it is stored for 2–6 hours during the first stages of processing. Complete digestion and

nutrient absorption occur in the small intestine over a period of 5–6 hours. Processing is completed in the large intestine, and undigested material is expelled through the anus as feces.

Animation

Overview of the Human Digestive System

image

image

Saliva is a complex mixture of materials with a number of vital functions. One major component is **mucus**, a viscous mixture of water, salts, cells, and slippery glycoproteins (carbohydrate-protein complexes). Mucus lubricates food for easier swallowing, protects the gums against abrasion, and facilitates taste and smell. Saliva also contains buffers, which help prevent tooth decay by neutralizing acid, and antimicrobial agents (such as lysozyme; see Figure 5.16), which protect against bacteria that enter the mouth with food.

The most abundant enzyme in saliva is **amylase**, which breaks down starch (a glucose polymer from plants) and glycogen (a glucose polymer from animals). While very little digestion occurs in the mouth, the action of chewing mixes amylase throughout the food where it can continue its digestive actions farther along the alimentary canal.

The tongue also has important roles in food processing. Taste buds on the tongue help evaluate ingested material, selecting which foods should be processed further and then enabling their passage. (See Concept 50.4 for a discussion of the sense of taste.) Once chewing begins, tongue movements manipulate the mixture of saliva and food, helping shape it into a ball called a *bolus***(Figure 41.9)**. During swallowing, the tongue provides further assistance, pushing the bolus to the back of the oral cavity and into the pharynx.

Figure 41.9

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Intersection of the human airway and digestive tract.
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The intersection of the human airway and digestive tract with a. trachea open and b. esophagus open.

In humans, the pharynx connects to the trachea and the esophagus. ***(a)*** *At most times, a contracted sphincter seals off the esophagus while the trachea remains open.* ***(b)*** *When a food bolus arrives at the pharynx, the swallowing reflex is triggered. Movement of the larynx, the upper part of the airway, tips a flap of tissue called the epiglottis down, preventing food from entering the trachea. At the same time, the esophageal sphincter relaxes, allowing the bolus to pass into the esophagus. The trachea then reopens, and peristaltic contractions of the esophagus move the bolus to the stomach.*

The intersection of the human airway and digestive tract with, a, trachea open and, b, esophagus open. Order of structures is: mouth with tongue and epiglottis (flap) at back, followed by the pharynx, which branches to, 1, the glottis and trachea; or 2, the esophageal sphincter and then the esophagus.

Part a. Trachea, leading to lungs, is open. Food bolus is in mouth. Epiglottitis is up. The esophageal sphincter at the bottom of the pharynx is closed, cutting off the route to the esophagus.

Part b. Esophagus is open. Epiglottis is down and glottis is up and closed. Food bolus is entering pharynx. Esophageal sphincter is relaxed, allowing food to pass into esophagus.

VISUAL SKILLS *If you laugh while drinking water, the liquid may be ejected from your nostrils. Use this diagram to explain why this happens, taking into account that laughing involves exhaling.*

*For suggested answer, see*Appendix A.**

Each bolus of food is received by the **pharynx**, or throat region, which leads to two passageways: the esophagus and the trachea. The **esophagus** is a muscular tube that connects to the stomach; the trachea (windpipe) leads to the lungs. Swallowing must therefore be carefully choreographed to keep food and liquids from entering the trachea and causing choking, a blockage of the trachea. The resulting lack of airflow into the lungs can be fatal if the material is not dislodged by vigorous coughing, a series of back slaps, or a forced upward thrust of the diaphragm (the Heimlich maneuver).

Within the esophagus, food is pushed along by **peristalsis**, alternating waves of smooth muscle contraction and relaxation. Upon reaching the end of the esophagus, the bolus encounters a **sphincter**, a ringlike valve of muscle **(Figure 41.10)**. Acting like a drawstring, the sphincter regulates passage of the ingested food into the next compartment, the stomach.

Figure 41.10

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The stomach and its secretions.
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1, Interior surface of the stomach, with cell types and functions detailed, and 2, outline of the production of gastric juice.

Parts are as follows.

1. Interior surface of stomach. The interior surface of the stomach wall is highly folded and dotted with pits leading into tubular gastric glands. Stomach wall is lined with simple columnar epithelium cells. Gastric glands. The gastric glands have three types of cells that secrete different components of the gastric juice: mucous cells, chief cells, and parietal cells. Mucous cells secrete mucous, which lubricates and protects the cells lining the stomach. Chief cells secrete pepsinogen,

an inactive form of the digestive enzyme pepsin. Parietal cells produce the components of hydrochloric acid (H C l).

2. The production of gastric juice. A close look at interior stomach surface cells reveals a single layer of cells that consists of chief cells and larger parietal cells, here curved into a U shape with the lumen of the stomach at the center. 1. Pepsinogen and H C l are introduced into the lumen of the stomach when parietal cells secrete H plus and C l minus ions and chief cells secrete pepsinogen. 2. H C l converts pepsinogen to pepsin (active enzyme). 3. Pepsin then activates more pepsinogen, starting a chain reaction. Pepsin begins the chemical digestion of proteins.

Figure 41.10 Walkthrough

The stomach and its secretions

Watch an explanation of this figure in the Figure 41.10 Walkthrough: The stomach and its secretions.

image

image

Digestion in the Stomach

The **stomach**, which is located just below the diaphragm, has two major roles in digestion. The first is storage. With a very elastic wall and accordion-like folds, the stomach can stretch to accommodate about 2 L of food and fluid. The second major function is to process food into a liquid suspension. The stomach secretes a digestive fluid called **gastric juice** and mixes it with the food through a churning action. This mixture of ingested food and gastric juice is called **chyme**.

Chemical Digestion in the Stomach

Two components of gastric juice help liquefy food in the stomach. First, hydrochloric acid (HCl) disrupts the extracellular matrix that binds cells together in meat and plant material. The concentration of HCl is so high that the pH of gastric juice is about 2, acidic enough to dissolve iron nails (and to kill most bacteria). This low pH denatures (unfolds) proteins in food, increasing exposure of their peptide bonds. The exposed bonds are then attacked by the second component of gastric juice—a **protease**, or protein-digesting enzyme, called **pepsin**. Unlike most enzymes, pepsin is adapted to work best in a very acidic environment. By breaking peptide bonds, it cleaves proteins into smaller polypeptides and further exposes the contents of ingested tissues.

Two types of cells in the gastric glands of the stomach produce the components of gastric juice (see Figure 41.10). *Parietal cells* use an ATP-driven pump to expel hydrogen ions into the lumen. At the same time, chloride ions diffuse into the lumen through specific membrane channels of the parietal cells. It is therefore only within the lumen that hydrogen and chloride ions combine to form HCl. Meanwhile, *chief cells* release pepsin into the lumen in an inactive form called **pepsinogen**. HCl converts pepsinogen to active pepsin by clipping off a small portion of the molecule and exposing its active site. Through these processes, both HCl and pepsin form in the lumen (cavity) of the stomach, not within the cells of the gastric glands. As a result, the parietal and chief cells produce gastric juice but are not digested from within by its components.

After hydrochloric acid converts a small amount of pepsinogen to pepsin, pepsin itself helps activate the remaining pepsinogen. Pepsin, like HCl, can clip pepsinogen to expose the enzyme's active site. This generates more pepsin, which activates more pepsinogen. This series of events is an example of positive feedback (see Concept 40.2).

Why don't HCl and pepsin eat through the lining of the stomach? For one thing, mucus secreted by cells in gastric glands protects against self-digestion (see Figure 41.10). In addition, cell division adds a new epithelial layer every three days on average, replacing cells before the lining is fully eroded by digestive juices.

Stomach Dynamics

The breakdown of food by gastric juices is enhanced by muscular activity of the stomach. The coordinated series of muscle contractions and relaxations that we call "churning" mixes the stomach contents about every 20 seconds. Churning facilitates the action of HCl and pepsin by bringing all of the food into contact with the gastric juices secreted by the lining of the stomach. As a result, what began as a recently swallowed meal becomes the acidic, nutrient-rich broth known as chyme.

Contractions of stomach muscles also help move material through the alimentary canal. In particular, peristaltic contractions typically move the contents of the stomach into the small intestine within 2–6 hours after a meal. The sphincter located where the stomach opens to the small intestine helps regulate passage into the small intestine, allowing only one squirt of chyme to enter at a time.

Occasionally, the sphincter at the top of the stomach allows a movement, or flux, of chyme from the stomach back into the lower end of the esophagus. The painful irritation of the esophagus that results from this process of acid reflux is commonly called "heartburn."

Digestion in the Small Intestine

Although there is some chemical digestion in the oral cavity and stomach, most enzymatic hydrolysis of macromolecules from food occurs in the small intestine **(Figure 41.11)**. This organ's name refers to its small diameter compared to the large intestine, not to its length. The **small intestine** is in fact the alimentary canal's longest compartment—over 6m 20feet long in humans. The first 25 cm 10inches or so of the small intestine forms the **duodenum**. It is here that chyme from the stomach mixes with digestive juices from the pancreas, liver, and gallbladder, as well as from gland cells of the intestinal wall itself. As you will read in Concept 41.5, hormones released by the stomach and duodenum control the digestive secretions into the alimentary canal.

Figure 41.11

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Chemical digestion in the human digestive system.
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A table provides the sites for the digestion of carbohydrates, proteins, nucleic acids, and fats.

Chemical breakdown of nutrients occurs predominantly in the small intestine, making use of digestive enzymes specific to each class of nutrients.

Nutrient Class	Structure	Substrates	Enzymes	Products
Carbohydrate digestion	Oral cavity, pharynx, esophagus	Polysaccharides (starch, glycogen)	Salivary amylase	Smaller poly-saccharides, maltose
Carbohydrate Digestion	Small intestine (enzymes from pancreas)	Smaller polysaccharides	Pancreatic amylases	Disaccharides
Carbohydrate Digestion	Small intestine (enzymes from intestinal epithelium)	Disaccharides	Disaccharidases	Monosaccharides
Protein Digestion	Stomach	Proteins	Pepsin	Small polypeptides
Protein Digestion	Small intestine (enzymes from pancreas)	Small polypeptides	Pancreatic trypsin and chymotrypsin (These protein-digesting enzymes,	Smaller polypeptides

			or proteases, cleave bonds adjacent to certain amino acids).	
Protein Digestion	Small intestine (enzymes from intestinal epithelium)	Small peptides	Pancreatic carboxypeptidase	Smaller peptides, amino acids
Protein Digestion	Small intestine (enzymes from intestinal epithelium)	Small peptides	Dipeptidases, carboxypeptidase, aminopeptidase. (These proteases each split off one amino acid at a time from a dipeptide or polypeptide).	amino acids
Nucleic Acid Digestion	Small intestine (enzymes from pancreas)	DNA, RNA	Pancreatic nucleases	Nucleotides
Nucleic Acid	Small intestine (enzymes from intestinal epithelium)	Nucleotides	Nucleotidases	Nucleosides
Nucleic Acid	Small intestine (enzymes from intestinal epithelium)	Nucleosides	Nucleosidases and	Nitrogenous bases, sugars, phosphates

Pepsin is resistant to the denaturing effect of low pH *and is thus adapted to the environment of the stomach. Thinking about the small intestine, describe a different adaptation common to the digestive enzymes in that compartment.*

*For suggested answer, see*Appendix A.**

Animation

Digestive System Function

image

The arrival of chyme in the duodenum triggers release of the hormone secretin, which stimulates the **pancreas** to secrete bicarbonate. Bicarbonate neutralizes the acidity of chyme and acts as a buffer for chemical digestion in the small intestine. The pancreas also secretes numerous digestive enzymes into the small intestine. These include the proteases trypsin and chymotrypsin, which are produced in inactive forms. In a chain reaction similar to that for pepsinogen, they are activated when safely located in the lumen of the duodenum.

The epithelial lining of the duodenum is the source of additional digestive enzymes. Some are secreted into the lumen of the duodenum, whereas others are bound to the surface of epithelial cells. Together with the enzymes from the pancreas, they complete most digestion in the duodenum.

Fats present a particular challenge for digestion. Insoluble in water, they form large globules that cannot be attacked efficiently by digestive enzymes. In humans and other vertebrates, fat digestion is facilitated by bile salts, which act as emulsifiers (detergents) that break apart fat and lipid globules. Bile salts are a major component of **bile**, a secretion of the **liver** that is stored and concentrated in the **gallbladder**.

Bile production is metabolically linked to another vital liver function: the destruction of red blood cells that are no longer fully functional. Pigments released during red blood cell disassembly are incorporated into bile pigments, which are eliminated from the body with the feces. In some liver and blood disorders, bile pigments accumulate in the skin, resulting in a yellowing called jaundice.

Absorption in the Small Intestine

With digestion largely complete, the contents of the duodenum move into the *jejunum* and *ileum*, the remaining regions of the small intestine. There, nutrient absorption occurs across the lining of the intestine **(Figure 41.12)**. Large folds in the lining encircle the intestine and are studded with finger-shaped projections called **villi** (singular, *villus*). Within the villi, each epithelial cell has many microscopic projections, or **microvilli**, that face the intestinal lumen. These epithelial microvilli have a brush-like appearance that is reflected in the name *brush border*. Together, the folds, villi, and microvilli have a surface area of 200–300 m2, roughly the size of a tennis court. This enormous surface area is an evolutionary adaptation that greatly increases the rate of nutrient absorption (see Figure 33.9 for more discussion and examples of maximizing surface area in diverse organisms).

Figure 41.12

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Nutrient absorption in the small intestine.
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Nutrients absorbed by villi in intestinal wall, with a closer look at how nutrients move from the intestinal lumen into epithelial cells and across the basal surface towards the capillaries.

Water-soluble nutrients, such as amino acids and sugars, enter the bloodstream, whereas fats are transported to the lymphatic system.

Long Description: [DO NOT DELETE THIS LINE]

The structure of the small intestine is a heavily folded tube. On the outer surface are muscle layers and blood vessels, including veins carrying blood to the liver. The inner small intestinal wall is lined with brush-like villi arranged in large circular folds; nutrients are absorbed by the villi. A closer look at the villi reveals finger-shaped projections covered with blood capillaries and lacteals that run into larger blood and lymph vessels at the finger bases. Epithelial cells line each finger projection. Nutrients are absorbed into the capillaries. A closer look at the absorption of nutrients in the villi reveals that nutrients are absorbed from the lumen of the small intestine into epithelial cells lining the villi, and the nutrients pass through the epithelial cells and out the basal surface to the capillary.

Tapeworms sometimes infect the human alimentary canal, anchoring themselves to the wall of the small intestine. Based on how digestion is compartmentalized along the mammalian alimentary canal, what digestive functions would you expect these parasites to have?

*For suggested answer, see*Appendix A.**

BioFlix® Animation

Nutrient Transport Across Membranes

image

Depending on the nutrient, transport across the epithelial cells can be passive or active (see Concepts 7.3 and 7.4). The sugar fructose, for example, moves by facilitated diffusion down its concentration gradient from the lumen of the small intestine into the epithelial cells. From there, fructose exits the basal surface and is absorbed into microscopic blood vessels, or capillaries, at the core of each villus. Other nutrients, including amino acids, small peptides, vitamins, and most glucose molecules, enter the epithelial cells of the villus by being pumped against concentration gradients. This active transport allows much more absorption of those nutrients than would be possible with passive diffusion alone.

The capillaries and veins that carry nutrient-rich blood away from the villi converge into the **hepatic portal vein**, a blood vessel that leads directly to the liver. From the liver, blood travels to the heart and then to other tissues and organs. This arrangement serves two major functions. First, it allows the liver to regulate the distribution of nutrients to the rest of the body. Because the liver converts many organic nutrients to

different forms for use elsewhere, blood leaving the liver may have a very different nutrient balance than the blood that entered. Second, the arrangement allows the liver to remove toxic substances before they can circulate broadly. The liver is the primary site for detoxifying many organic molecules foreign to the body, such as drugs, and certain metabolic waste products.

Although many nutrients leave the small intestine through the bloodstream and pass through the liver for processing, some products of fat (triglyceride, also known as triacylglycerol) digestion take a different path **(Figure 41.13)**. Hydrolysis of a fat by lipase in the small intestine generates fatty acids and a monoglyceride (glycerol joined to a fatty acid). These products are absorbed by epithelial cells and recombined into triglycerides. They are then coated with phospholipids, cholesterol, and proteins, forming globules called **chylomicrons** that are too large to enter blood vessels.

Figure 41.13

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Digestion and absorption of fats.
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Digestion and absorption of fats by the small intestine includes the enzymatic hydrolysis of triglycerides in the lumen and the formation of chylomicrons inside the epithelial cells.

Fats, which are insoluble in water, are broken down in the lumen of the small intestine, reassembled in epithelial cells, and then transported in water-soluble globules called chylomicrons. The chylomicrons enter the lymph via narrow vessels called lacteals and are later transferred to the blood in large veins leading to the liver and heart.

Steps are as follows.

1. In the lumen of the small intestine, bile salts break up large fat globules into fat droplets, increasing exposure of triglycerides (fat molecules) on the surface to hydrolysis.
2. During enzymatic hydrolysis, the enzyme lipase breaks the exposed triglycerides down to fatty acids and monoglycerides.
3. After diffusing into epithelial cells, monoglycerides and fatty acids are re-formed into triglycerides. (Some glycerol and fatty acids pass directly into capillaries.)
4. The triglycerides are incorporated into particles called chylomicrons. Phospholipids and proteins on the surface make chylomicrons water-soluble.
5. Chylomicrons leave epithelial cells by exocytosis and enter lacteals, where they are carried away by the lymph and later pass into large veins that lead directly to the heart.

VISUAL SKILLS *Arrows at two locations in this figure indicate movement of materials between the epithelial cell and its surroundings. Identify and circle these arrows. At either of these locations, do the movements represented require an input of energy? Explain.*

*For suggested answer, see*Appendix A.**

HHMI BioInteractive Animation

The Fate of Fat

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In exiting the small intestine, chylomicrons first enter a **lacteal**, a vessel at the core of each villus that has large gaps between adjacent cells (see Figure 41.13, step 5). Lacteals are part of the vertebrate lymphatic system, which is a network of vessels filled with a clear fluid called lymph. Starting at the lacteals, lymph containing the chylomicrons passes into the larger vessels of the lymphatic system and eventually into large veins that return the blood directly to the heart.

In addition to absorbing nutrients, the small intestine recovers water and ions. Each day a person consumes about 2 L of water and secretes another 7 L in digestive juices into the alimentary canal. Typically all but 0.1 L of the water is reabsorbed, mostly in the small intestine. There is no mechanism for active transport of water. Instead, water is reabsorbed by osmosis when sodium and other ions are pumped out of the lumen of the small intestine.

Processing in the Large Intestine

The alimentary canal ends with the **large intestine**, which includes the colon, cecum, and rectum. The small intestine connects to the large intestine at a T-shaped junction **(Figure 41.14)**. One arm of the T is the 1.5-m-long **colon**, which leads to the rectum and anus. The other arm is a pouch called the **cecum**. In animals that eat large amounts of plant material, the cecum has an important role in fermenting ingested material. In humans, the cecum is small and has an **appendix**, a finger-shaped extension that acts as a reservoir for symbiotic microorganisms, which are discussed in Concept 41.4.

Figure 41.14

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Junction of the small and large intestines.
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Junction of small and large intestines with small intestine connecting at point below the ascending portion of colon and above the cecum and appendix.

The colon completes the recovery of water that began in the small intestine. What remain are the **feces**, the wastes of the digestive system, which become increasingly solid as they are moved along the colon by peristalsis. It takes approximately 12–24 hours for material to travel the length of the colon. The undigested material in feces includes cellulose fiber. Although it provides no caloric value (energy) to humans, it helps move food along the alimentary canal.

If the lining of the colon is irritated—by a viral or bacterial infection, for instance—less water than normal may be reabsorbed, resulting in diarrhea. The opposite problem, constipation, occurs when the feces move along the colon too slowly. Too much water is reabsorbed, and the feces become compacted.

The community of bacteria living on unabsorbed organic material in the human colon contributes about one-third of the dry weight of feces. As by-products of their metabolism, many colon bacteria generate gases, including methane and hydrogen sulfide, the latter of which has an offensive odor. These gases and ingested air are expelled through the anus.

The terminal portion of the large intestine is the **rectum**, where the feces are stored before elimination. Two sphincters separate the rectum and the anus; the inner one is involuntary and the outer one is voluntary. Periodically, strong contractions of the colon create an urge to defecate. Because filling of the stomach triggers a reflex that increases the rate of contractions in the colon, the urge to defecate often follows a meal.

Having followed a meal through the alimentary canal, we'll look next at some adaptations of this general digestive plan in different animals.

Concept 41.4: Evolutionary adaptations of vertebrate digestive systems correlate with diet

EVOLUTION The digestive systems of vertebrates are varied, with many adaptations linked to an animal's diet. To highlight how form fits function, we'll examine a few of them.

Dental Adaptations

Dentition, an animal's assortment of teeth, is one example of structural variation reflecting diet **(Figure 41.15)**. The evolutionary adaptation of teeth for processing different kinds of food is one of the major

reasons that mammals have been so successful. For example, a sea otter uses its sharp canine teeth to tear apart prey such as crabs and its slightly rounded molars to crush their shells. Nonmammalian vertebrates generally have less specialized dentition, but there are interesting exceptions. Venomous snakes, such as rattlesnakes, have fangs, modified teeth that inject venom into prey. Some fangs are hollow, like syringes, whereas others drip the toxin along grooves on the surfaces of the teeth.

Figure 41.15

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Dentition and diet.
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Arrangement of incisors, canines, premolars and molars in carnivores, herbivores and omnivores. Dentition patterns correspond to diet type.

Arrangement of incisors, canines, premolars and molars in carnivores, herbivores and omnivores. Teeth are described from front to back.

Carnivore. Carnivores, such as members of the dog and cat families, generally have large, pointed incisors; canines are prominent and also pointed. Both can be used to kill prey and rip or cut away pieces of flesh. The jagged premolars and molars cut and shred food.

Herbivore. Herbivores, such as horses and deer, generally have incisors and canines that are modified for biting off pieces of vegetation. In the skull shown there is a gap in the elongated jaw between the canines and premolars. Herbivores usually have premolars and molars with broad, ridged surfaces that grind tough plant material.

Omnivore. As omnivores, humans are adapted to eating both plants and meat. Adults have 32 teeth. From front to back along either side of the mouth are four blade-like incisors for biting, a pair of pointed canines for tearing, four premolars for grinding, and six molars with a notched top surface for crushing.

Stomach and Intestinal Adaptations

Evolutionary adaptations to differences in diet are sometimes apparent as variations in the dimensions of digestive organs. For example, large, expandable stomachs are common in carnivorous vertebrates, which may wait a long time between meals and must eat as much as they can when they do catch prey. An expandable stomach enables a rock python to ingest a whole gazelle (see Figure 41.5) and a 200-kg African lion to consume 40kg of meat in one meal!

Adaptation is also apparent in the length of the digestive system in different vertebrates. In general, herbivores and omnivores have longer alimentary canals relative to their body size than do carnivores. Plant

matter is more difficult to digest than meat because it contains cell walls. A longer digestive tract furnishes more time for digestion and more surface area for nutrient absorption. As an example, consider the lion and koala in **Figure 41.16**. The koala has much longer intestines, relative to its size, enhancing the processing of fibrous, protein-poor eucalyptus leaves from which it obtains nearly all of its nutrients and water.

Figure 41.16

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    The alimentary canals of a carnivore (lion) and herbivore
(koala).
```

Comparison of alimentary canals of a coyote and koala. The alimentary canal of the koala is much longer, with a greatly increased cecum and colon as compared to the coyote.

The relatively short digestive tract of the lion is sufficient for digesting meat and absorbing its nutrients. In contrast, the koala's long alimentary canal is specialized for digesting eucalyptus leaves. Extensive chewing chops the leaves into tiny pieces, increasing exposure to digestive juices. In the long cecum and the upper portion of the colon, symbiotic bacteria further digest the shredded leaves, releasing nutrients that the koala can absorb.

Comparison of the alimentary canals of a coyote (carnivore) and koala (herbivore). The stomachs of both animals are similar. Following the stomach is the small intestine, and then the cecum for both animals. The coyote cecum is a small intestinal extension, but the koala cecum is a long, coiled loop. Likewise, the colon (large intestine) of the coyote is relatively short as compared to the koala, which has a much longer colon.

Mutualistic Adaptations

An estimated 10–100 trillion bacteria live in the human digestive system. The coexistence of humans and many intestinal bacteria is an example of mutualism, an interaction between two species that benefits both of them (see Concept 54.1). For example, some intestinal bacteria produce vitamins, such as vitamin K, biotin, and folic acid, which are absorbed into the blood, supplementing our dietary intake. Intestinal bacteria also regulate the development of the intestinal epithelium and the function of the innate immune system. The bacteria in turn receive a steady supply of nutrients and a stable host environment.

Recently, we have greatly expanded our knowledge of the **microbiome**, the collection of microorganisms living in and on the body, along with their genetic material. To study the microbiome, scientists are using a DNA sequencing approach based on the polymerase chain reaction (PCR, see Figure 20.8). To date they have found more than 400 bacterial species in the human digestive tract, a far greater number than had

been identified through approaches relying on laboratory culture and characterization. Furthermore, researchers have found significant differences in the microbiome associated with diet, disease, and age **(Figure 41.17)**.

Figure 41.17

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Variation in human gut microbiome at different life stages.
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Comparison of the relative proportions of firmicutes, bacteroidetes, actinobacteria, proteobacteria, and other bacterial phyla in human infants, toddlers, adults, and elderly people.

By copying and sequencing bacterial DNA in samples obtained from intestinal tracts of humans of different ages, researchers characterized the bacterial community that makes up the human gut and how it changes with age.

In a human infant on breastmilk, the gut microbiome consists of about 50% actinobacteria, 25% proteobacteria, 13% firmicutes, 10% bacteroidetes, and 2% other bacterial phyla. In a human infant on formula, the gut microbiome consists of about 45% actinobacteria, 45% firmicutes, and 10% proteobacteria. In a human infant on solid food, the gut microbiome consists of about 45% bacteroidetes, 33% firmicutes, 20% other bacterial phyla, and 2% proteobacteria. In a human toddler who is healthy, the gut microbiome consists of about 45% bacteroidetes, 33% firmicutes, 20% other bacterial phyla, and 2% proteobacteria. In a human toddler on antibiotic treatment, the gut microbiome consists of about 78% bacteroidetes, 21% firmicutes, and 1% proteobacteria. In a human toddler who is malnourished, the gut microbiome consists of about 40% proteobacteria, 33% firmicutes, 20% bacteroidetes, 5% other bacterial phyla, and 2% actinobacteria. In a healthy human adult, the gut microbiome consists of about 75% firmicutes, 19% bacteroidetes, 3% actinobacteria, 2% proteobacteria, and 1% other bacterial phyla. In an obese human adult, the gut microbiome consists of about 65% firmicutes, 15% bacteroidetes, 12% proteobacteria, and 8% actinobacteria. In an elderly human aged 65-80 years old, the gut microbiome consists of about 78% firmicutes, 14% bacteroidetes, 4% actinobacteria, 3% proteobacteria, and 1% other bacterial phyla. In an elderly human over 100 years old, the gut microbiome consists of about 60% firmicutes, 20% actinobacteria, 18% bacteroidetes, 1% proteobacteria, and 1% other bacterial phyla. All percentages are approximate.

INTERPRET THE DATA *Using data displayed in**Figures 41.17**and*41.18*, compare the relative abundance of Actinobacteria in the microbiome of a healthy adult's intestinal tract to that in a healthy stomach. Suggest a possible explanation for why the microbiome composition in the two organs is different even though the intestine and stomach are directly connected.*

One example of the power of microbiome studies comes from research on gastric ulcers, a disease that damages the stomach lining. Experiments demonstrating that ulcers are caused by infection with the acid-tolerant bacterium *Helicobacter pylori* and can be cured with antibiotics earned Australian researchers Barry Marshall and Robin Warren a Nobel Prize in 2005.

Recently, scientists analyzed the microbiome in samples from human stomachs to learn how *H. pylori* infection leads to ulcers. Their findings were dramatic: *H. pylori* infection led to the near-complete elimination of all other bacterial species commonly found in the stomach **(Figure 41.18)**.

Figure 41.18

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The stomach microbiome and stomach health.
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Bar chart of percent of sampled stomach bacteria by bacterial phylum in individuals uninfected and infected with H. pylori. When infected, H. pylori makes up over 95% of bacteria present.

Bar chart of percent of sampled stomach bacteria by bacterial phylum in individuals uninfected and infected with H. pylori. The horizontal axis is bacterial phylum with a bar for each phylum, and the vertical axis is percent of sampled stomach bacteria, ranging from 0 to 100 with increments every 20 percent. Uninfected individuals have bacteria present from the phyla (in decreasing order of percent sampled) Actinobacteria, Firmicutes, Bacteroidetes, Proteobacteria and Fuscobacteria; infected individuals have more than 95% H. pylori and only negligible occurrences of other phyla. H. pylori are rod-shaped bacteria with flagella.

In samples from individuals infected with *Helicobacter pylori*, more than 95% of the sequences were from that species, which belongs to the phylum Proteobacteria. The stomach microbiome in uninfected individuals was much more diverse.

Studies on the microbiome have already led to therapies for intestinal infections by antibiotic-resistant pathogens, a major public health problem. One such therapy is fecal microbial transplantation, in which the microbiome of a healthy individual is introduced into the patient's intestine. This approach has been used for otherwise untreatable diarrhea caused by the bacterium *Clostridium difficile* (**Figure 41.19)**. Such infections are most common after antibiotic treatment has killed off the resident microbiome. Clinical trials have shown the fecal transplantation approach to be effective, although there is a real risk of secondary infection.

Figure 41.19

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*Clostridium difficile.*
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A bacterial cell with many flagella.

Another treatment against antibiotic-resistant pathogens makes use of bacteriophages, viruses that infect bacteria but not human cells (see Concept 19.1). In 2017, bacteriophages engineered to kill a multidrug-resistant bacterial pathogen, *Acinetobacter baumannii*, were successfully used to restore health to a severely affected patient.

Interview

Interview with Steffanie Strathdee: Harnessing phages to fight deadly infections

Steffanie Strathdee.

Mutualistic Adaptations in Herbivores

Mutualistic relationships with microorganisms are also very important in herbivores. Herbivores get much of the chemical energy they need from the cellulose of plant cell walls, but, like other animals, herbivores do not produce enzymes that hydrolyze cellulose. Instead, many vertebrates (as well as termites, whose wooden diets consist largely of cellulose) host large populations of mutualistic bacteria and protists in fermentation chambers in their alimentary canals. These symbiotic microorganisms have enzymes that can digest cellulose to simple sugars and fatty acids that the animal can absorb. In many cases, the microorganisms also use the sugars from digested cellulose in the production of a variety of essential nutrients, such as vitamins and amino acids.

In horses, koalas, and elephants, symbiotic microorganisms are housed in a large cecum. In contrast, the hoatzin, an herbivorous bird found in South American rain forests, hosts microorganisms in a large, muscular crop (an esophageal pouch; see Figure 41.7). Hard ridges in the wall of the crop grind plant leaves into small fragments, and the microorganisms break down cellulose.

In rabbits and some rodents, mutualistic bacteria live in the large intestine as well as the cecum. Since most nutrients are absorbed in the small intestine, nourishing by-products of fermentation by bacteria in the large intestine are initially lost with the feces. Rabbits and rodents recover these nutrients by *coprophagy* (from the Greek, meaning “dung eating”), feeding on some of their feces and then passing the food through the alimentary canal a second time. The familiar rabbit “pellets,” which are not reingested, are the feces eliminated after food has passed through the digestive tract twice.

The most elaborate adaptations for an herbivorous diet have evolved in the animals called ***ruminants***, the cud-chewing animals that include deer, sheep, and cattle **(Figure 41.20)**.

Figure 41.20

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Ruminant digestion.
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Path of food (grass) through cow is mouth, esophagus, rumen, reticulum (with regurgitation to mouth for re-chewing cud), omasum, abomasum, intestine.

The stomach of a cow, a ruminant, has four chambers. ***1*** *Chewed food first enters the rumen and reticulum, where mutualistic microorganisms digest cellulose in the plant material.* ***2*** *Periodically, the cow regurgitates and rechews "cud" from the reticulum, further breaking down fibers and thereby enhancing microbial action.* ***3*** *The reswallowed cud passes to the omasum, where some water is removed.* ***4*** *It then passes to the abomasum for digestion by the cow's enzymes. In this way, the cow obtains significant nutrients from both the grass and the mutualistic microorganisms, which maintain a stable population in the rumen.*

Although we have focused our discussion on vertebrates, adaptations related to digestion are also widespread among other animals. Some of the most remarkable examples are the over 3-meter-long giant tubeworms **(Figure 41.21)** that live at pressures as high as 260 atmospheres around deep-sea hydrothermal vents (see Figure 52.16). These worms have no mouth or digestive system. Instead, they obtain all of their energy and nutrients from mutualistic bacteria that live within their bodies. The bacteria carry out chemoautotrophy (see Concept 27.3) using the carbon dioxide, oxygen, hydrogen sulfide, and nitrate available at the vents. Thus, for invertebrates and vertebrates alike, the evolution of mutualistic relationships with symbiotic microorganisms is an adaptation that expands the sources of nutrition available to animals.

Figure 41.21

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Giant tubeworm, an animal without a digestive system.
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Giant tubeworm with highly vascularized red plume at free end.

Having examined how animals optimize their extraction of nutrients from food, we'll next turn to the challenge of balancing the use of these nutrients.

Concept 41.5 Feedback circuits regulate digestion, energy storage, and appetite

To complete our consideration of animal nutrition, we'll explore the ways that obtaining and using nutrients are matched to an animal's circumstances and need for energy.

Regulation of Digestion

Many animals face long gaps between meals. Under such circumstances, there is no need for their digestive systems to be active continuously. Instead, processing is activated stepwise. As food reaches each new compartment, it triggers the secretion of digestive juices for the next stage of processing. Muscular contractions then move the contents farther along the canal. For example, you learned earlier that nervous reflexes stimulate the release of saliva when food enters the oral cavity and orchestrate swallowing when a bolus of food reaches the pharynx. Similarly, the arrival of food in the stomach triggers churning and the release of gastric juices. These events, as well as movement through the small and large intestines, are regulated by the *enteric nervous system*, a network of neurons dedicated to the digestive organs.

The endocrine system also plays a critical role in controlling digestion. As described in **Figure 41.22**, a series of hormones released by the stomach and duodenum help ensure that digestive secretions are present only when needed. Like all hormones, they are transported through the bloodstream. This is true even for the hormone gastrin, which is secreted by the stomach and targets that same organ.

Figure 41.22

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Hormonal control of digestion.
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Inhibition and stimulation highlighted in the hormonal control of digestion, which involves the stomach, gall bladder and pancreas, as well as various hormones.

Steps are as follows.

1. Diagram of large liver connected to small gallbladder, connected to pancreas (beneath stomach) and duodenum of small intestine. Food travels to stomach (connected to duodenum). As food arrives at the stomach, it stretches the stomach walls, triggering release of the hormone gastrin.

Gastrin circulates via the bloodstream back to the stomach, where it stimulates production of gastric juices (stimulation).

2. Chyme, an acidic mixture of partially digested food, eventually passes from the stomach to the duodenum. The duodenum responds to amino acids or fatty acids in the chyme by releasing the digestive hormones cholecystokinin and secretin. Cholecystokinin (CCK) stimulates the release of digestive enzymes from the pancreas and of bile from the gall bladder (stimulation). Secretin stimulates the pancreas to release bicarbonate (H C O 3 minus), which neutralizes chyme (stimulation).

3. If the chyme is rich in fats, the high levels of secretin and CCK released act on the stomach to inhibit peristalsis and secretion of gastric juices, thereby slowing digestion (inhibition).

Regulation of Energy Storage

When an animal takes in more energy-rich molecules than it needs for metabolism and activity, it stores the excess energy (see Concept 40.4). In humans, energy can be stored in liver and muscle as glycogen, a polysaccharide made up of many glucose units (see Figure 5.6b), or in adipose cells as triglycerides (fat). Glycogen stores are limited, but fat stores are not. A large grizzly bear can easily gain 75kg (165 lb) of fat as it prepares for hibernation.

At times when fewer calories are taken in than are expended—perhaps because of sustained heavy exercise or lack of food—muscle cells draw on their stored glycogen, and liver and adipose cells release glucose and fatty acids into the blood for use by other body cells. Fats are especially rich in energy; oxidizing a gram of fat liberates about twice the energy liberated from a gram of carbohydrate or protein. For this reason, ******adipose tissue** provides the most space-efficient way for the body to store large amounts of energy. Most healthy people have enough stored fat to sustain them through several weeks without food.

Glucose Homeostasis

The synthesis and breakdown of glycogen are central not only to energy storage, but also to maintaining metabolic balance through glucose homeostasis. In humans, the normal range for the concentration of glucose in the blood is 70–110 mg/100 mL. Because glucose is a major fuel for cellular respiration and a key source of carbon skeletons for biosynthesis, maintaining blood glucose concentrations near this set point is essential.

Glucose homeostasis relies predominantly on the antagonistic (opposing) effects of two hormones, insulin and glucagon **(Figure 41.23)**. When the blood glucose level rises above the set point, increased secretion of

insulin triggers the uptake of glucose from the blood into body cells, decreasing the blood glucose concentration. When the blood glucose level drops below the set point, increased secretion of **glucagon** promotes the release of glucose into the blood from energy stores, such as liver glycogen, increasing the blood glucose concentration.

Figure 41.23

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Homeostatic regulation of cellular fuel.
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Flow diagram depicts the roles of hormones insulin and glucagon (both produced in the pancreas) in regulating blood glucose levels to normal (70 to 110 milligrams glucose per 100 milliliters).

After a meal is digested, glucose and other monomers are absorbed into the blood from the digestive tract. The human body regulates the use and storage of glucose, a major cellular fuel.

Flow diagram depicts the roles of the hormones insulin and glucagon in regulating blood glucose levels. Normal blood glucose is 70 to 110 milligrams glucose per 100 milliliters. Blood glucose level rises (such as after eating), and the beta cells of the pancreas secrete the hormone insulin into the blood. Insulin enhances the transport of glucose into body cells and stimulates the liver to store glucose as glycogen, resulting in the fall of blood glucose levels back to normal. From normal, blood glucose level can fall (such as after fasting), resulting in the alpha cells of the pancreas secreting the hormone glucagon into the blood. Glucagon promotes the breakdown of glycogen in the liver and the release of glucose into the blood, which results in blood glucose level rising back to normal.

MAKE CONNECTIONS *What form of feedback control do these regulatory circuits reflect (see*Concept 40.2*)?**

*For suggested answer, see*Appendix A.**

BioFlix® Animation

Homeostasis: Regulating Blood Sugar

image

Play

image

Animation

Pancreatic Hormones Regulate Blood Glucose Level

image

image

The liver is a key site of action for insulin and glucagon. After a carbohydrate-rich meal, for example, insulin promotes biosynthesis of glycogen from glucose that enters the liver in the hepatic portal vein. Between meals, when the blood glucose concentration decreases, glucagon stimulates the liver to break down glycogen, convert amino acids and glycerol to glucose, and release glucose into the blood. Together, these opposing effects of insulin and glucagon ensure that arterial blood has a glucose concentration within the set point at nearly all times.

Insulin also acts on nearly all body cells to stimulate glucose uptake from blood. A major exception is brain cells, which can take up glucose whether or not insulin is present. This evolutionary adaptation ensures that the brain almost always has access to circulating fuel, even if supplies are low.

Glucagon and insulin are both produced in the pancreas. Clusters of endocrine cells called pancreatic islets are scattered throughout this organ. Each islet has *alpha cells*, which make glucagon, and *beta cells*, which make insulin. Like other hormones, insulin and glucagon enter the interstitial fluid and then the circulatory system.

Overall, hormone-secreting cells make up only 1–2% of the mass of the pancreas. Other cells in the pancreas produce and secrete bicarbonate ions and the digestive enzymes active in the small intestine (see Figure 41.11). These secretions are released into small ducts that empty into the pancreatic duct, which leads to the small intestine. Thus, the pancreas has functions in both the endocrine and digestive systems.

Diabetes Mellitus

In discussing the role of insulin and glucagon in glucose homeostasis, we have focused exclusively on a healthy metabolic state. However, a number of disorders can disrupt glucose homeostasis with potentially serious consequences, especially for the heart, blood vessels, eyes, and kidneys. The best known and most prevalent of these disorders is diabetes mellitus.

The disease **diabetes mellitus** is caused by a deficiency of insulin or a decreased response to insulin in target tissues. The blood glucose level rises, but cells are unable to take up enough glucose to meet metabolic needs. Instead, fat becomes the main substrate for cellular respiration. In severe cases, acidic metabolites formed during fat breakdown accumulate in the blood, threatening life by lowering blood pH and depleting sodium and potassium ions from the body.

In people with diabetes mellitus, the level of glucose in the blood may exceed the capacity of the kidneys to reabsorb this nutrient. Glucose that remains in the kidney filtrate is excreted. For this reason, the presence

of sugar in urine is one test for this disorder. The presence of excess sugar in both urine and blood is the basis for the name *diabetes mellitus* (from the Greek *diabainein*, to pass through, and *meli*, honey).

There are two main types of diabetes mellitus: type 1 and type 2. Each is marked by high blood glucose levels, but with very different causes.

Type 1 Diabetes

Also called insulin-dependent diabetes, *type 1 diabetes* is an autoimmune disorder in which the immune system destroys the beta cells of the pancreas. Type 1 diabetes, which usually appears during childhood, destroys the person's ability to produce insulin. Treatment consists of insulin injections, typically given multiple times daily. In the past, insulin was extracted from animal pancreases, but now human insulin can be readily obtained at reasonable cost from genetically engineered bacteria (see Figure 20.4). Stem cell research may someday provide a cure for type 1 diabetes by generating replacement beta cells that restore insulin production by the pancreas.

Type 2 Diabetes

Non-insulin-dependent diabetes, or *type 2 diabetes*, is characterized by a failure of target cells to respond normally to insulin. Insulin is produced, but target cells fail to take up glucose from the blood, and the blood glucose level remains elevated. Although heredity can play a role in type 2 diabetes, excess body weight and lack of exercise significantly increase the risk of developing this disorder. This form of diabetes generally appears after age 40, but even children can develop the disease, particularly if they are overweight and sedentary. More than 90% of people with diabetes have type 2. Many can control their blood glucose levels with regular exercise and a healthy diet; some require medications. Nevertheless, type 2 diabetes is a leading cause of death in the United States and a growing public health problem worldwide.

The resistance to insulin signaling in type 2 diabetes is sometimes due to a genetic defect in the insulin receptor or the insulin response pathway. In many cases, however, events in target cells suppress activity of an otherwise functional response pathway. One source of this suppression appears to be inflammatory signals generated by the innate immune system (see Concept 43.1). How obesity and inactivity relate to this suppression is being studied in both humans and laboratory animals.

BioFlix® Animation

Diabetes

image

Play

image

Regulation of Appetite and Consumption

Consuming more calories than the body needs for normal metabolism, or *overnourishment*, can lead to obesity, the excessive accumulation of fat. Obesity, in turn, contributes to a number of health problems, including type 2 diabetes, cancer of the colon and breast, and cardiovascular disease that can result in heart attacks and strokes. During the COVID-19 pandemic, obesity was identified as a major risk factor for COVID-19-associated hospitalization and death, particularly during the early waves of infection.

Researchers have discovered several homeostatic mechanisms that operate as feedback circuits controlling the storage and metabolism of fat. A network of neurons relays and integrates information from the digestive system to regulate secretion of hormones that regulate long-term and short-term appetite. The target for these hormones is a "satiety center" in the brain **(Figure 41.24)**. For example, *ghrelin*, a hormone secreted by the stomach wall, triggers feelings of hunger before meals. In contrast, both insulin and PYY, a hormone secreted by the small intestine after meals, suppress appetite. *Leptin*, a hormone produced by adipose (fat) tissue, also suppresses appetite and appears to play a major role in regulating body fat levels. In the **Scientific Skills Exercise**, you'll interpret data from an experiment studying genes that affect leptin production and function in mice.

Figure 41.24

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A few of the appetite-regulating hormones.
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Some hormones that act as appetite stimulants or suppressors described, including the hormones ghrelin, insulin, leptin and PYY.

Secreted by various organs and tissues, the hormones reach the brain via the bloodstream. These signals act on a region of the brain that in turn controls the "satiety center," which generates the nervous impulses that make us feel either hungry or satiated ("full"). The hormone ghrelin is an appetite stimulant; the other three hormones shown here are appetite suppressants.

Appetite Stimulant: Secreted by the stomach wall, ghrelin is one of the signals that act on the satiety center in the brain to trigger feelings of hunger as mealtimes approach. In dieters who lose weight, ghrelin levels increase, which may be one reason it's so hard to stay on a diet.

Appetite suppressants: 1. A rise in blood sugar level after a meal stimulates the pancreas to secrete insulin. In addition to its other functions, insulin suppresses appetite by acting on the satiety center in the brain. 2.

Produced by adipose (fat) tissue, leptin suppresses appetite. When the amount of body fat decreases, leptin levels falls, and appetite increases. 3. The hormone PYY, secreted by the small intestine after meals, acts as an appetite suppressant that counters the appetite stimulant ghrelin.

Scientific Skills Exercise

Interpreting Data from an Experiment with Genetic Mutants

AP Science Practices 2, 3, 4, 5

What Are the Roles of theob**and**db**Genes in Appetite Regulation?**A mutation that disrupts a physiological process is often used to study the normal function of the mutated gene. Ideally, researchers use a standard set of conditions and compare animals that differ genetically only in whether a particular gene is mutant (nonfunctional) or wild-type (normal). In this way, a difference in phenotype, the physiological property being measured, can be attributed to a difference in genotype, the presence or absence of the mutation. To study the role of specific genes in regulating appetite, researchers used laboratory animals with known mutations in those genes.

Mice in which recessive mutations inactivate both copies of either the ob gene or the db gene eat voraciously and grow much more massive than wild-type mice.

Wild type thin mouse beside obese mouse with mutations to the o b gene.

The mouse on the right is wild-type, whereas the obese mouse on the left has an inactivating mutation in both copies of the ob gene.

One hypothesis for the normal role of the ob and db genes is that they participate in a hormone pathway that suppresses appetite when caloric intake is sufficient. Before setting out to isolate the potential hormone, researchers explored this hypothesis genetically.

How the Experiment Was Done The researchers measured the mass of young mice of various genotypes (the “subjects”) and surgically linked the circulatory system of each one to that of another mouse. This procedure ensured that any factor circulating in the bloodstream of either mouse would be transferred to the other in the pair. After eight weeks, they again measured the mass of each subject mouse.

Data from the Experiment

A table lists genotype pairing and the average change in body mass of the subject.

blank	Genotype pairing, subject	Genotype pairing, paired with	Average change in body mass of subject in grams
a.	o b + 1 a b +; d b + 1 d b +	o b + 1 o b +; d b + 1 d b +	8.3
b	o b 1 o b (mutant); d b 1 d b +	o b 1 o b (mutant); d b + 1 d b +	38.7
c.	o b 1 o b (mutant); d b + 1 d b +	o b + 1 o b +; d b + 1 d b +	8.2
d.	o b 1 o b (mutant); d b + 1 d b +	o b + 1 o b +; d b 1 d b (mutant)	negative 14.9, see note

Note, due to pronounced weight loss and weakening, subjects in this pairing were remeasured after less than 8 week.

Data from D. L. Coleman, Effects of parabiosis of obese mice with diabetes and normal mice, *Diabetologia* 9:294–298 (1973).

INTERPRET THE DATA

- First, practice reading the genotype information given in the data table. For example, pairing (a) joined two mice that each had the wild-type version of both genes. Describe the two mice in pairing (b), pairing (c), and pairing (d). Explain how each pairing contributed to the experimental design.
- Compare the results observed for pairing (a) and pairing (b) in terms of phenotype. If the results had been identical for these two pairings, what would that outcome have implied about the experimental design?
- Compare the results observed for pairing (c) to those observed for pairing (b). Based on these results, does the ob+ gene product appear to promote or suppress appetite? Explain your answer.
- Describe the results observed for pairing (d). Note how these results differ from those for pairing (b). Suggest a hypothesis to explain this difference. How could you test your hypothesis using the kinds of mice in this study?

Instructors: A version of this Scientific Skills Exercise can be assigned in **Mastering Biology.**

Obtaining food, digesting it, and absorbing nutrients are part of the larger story of how animals fuel their activities. Provisioning the body also involves circulating nutrients, and using nutrients for metabolism

requires exchanging respiratory gases with the environment. These processes and the adaptations that facilitate such distribution and exchange are the focus of Chapter 42.